PROPERTIES P OF MORPHISMS OF SPACES ARE CHECKED ON ONE FPPF COVER OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 682-693 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 695-713, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 24 of queue_stacks-properties.json (pb=705, stmt 682-693, proof 695-713).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-property-spaces-too: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 650–658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let P be a property of morphisms of algebraic spaces as above. Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks representable by algebraic spaces. Let W be an algebraic space and let $W \to \mathcal{Y}$ be surjective, locally of finite presentation, and flat. Set $V = W \times_{\mathcal{Y}} \mathcal{X}$. Then*

$$(f \text{ has } P) \Leftrightarrow (\text{the projection } V \to W \text{ has } P).$$

Proof. The implication from left to right follows from Lemma lemma-property-spaces-too. Assume $V \to W$ has P. Let T be a scheme, and let $T \to \mathcal{Y}$ be a morphism. Consider the commutative diagram

$$\begin{array}{ccccc} T \times_{\mathcal{Y}} \mathcal{X} & \longleftarrow & T \times_{\mathcal{Y}} V & \longrightarrow & V \\ \downarrow & & \downarrow & & \downarrow \\ T & \longleftarrow & T \times_{\mathcal{Y}} W & \longrightarrow & W \end{array}$$

of algebraic spaces. The squares are cartesian. The bottom left morphism is a surjective, flat morphism which is locally of finite presentation, hence $\{T \times_{\mathcal{Y}} V \to T\}$

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is an fppf covering. Hence the fact that the right vertical arrow has property P implies that the left vertical arrow has property P. $\square$